Biochemistry Cumulative Examination

October 26 1996

1. Reversible protein phosphorylation is a critical mechanism for the regulation of protein activity. Surprisingly, only in three cases has the structural basis of phosphorylation induced functional modification been determined. The latest example, the dimeric protein yeast glycogen phosphorylase, was recently examined by comparing the structure of the unphosphorylated, inactive enzyme to the phosphorylated, active enzyme (Lin *et al.*, Science, 1996, **273**, 1539). In yeast glycogen phosphorylase a single, specific threonine is phosphorylated. What other amino acids, if any, are commonly phosphorylated in other proteins that are reversibly phosphorylated? (15 points)

2. Draw the structure of the phosphorylated version of a Gly-Thr-Gly tripeptide as it would appear in aqueous solution at pH 5. You need not show absolute stereochemistry but be sure to include all the correct atoms in the phosphate moiety and show charges. (15 points)

3. What is the generic name for enzymes that phosphorylate other proteins? (A single word) (10 points)

4. Yeast glycogen phosphorylase is an example of an allosteric enzyme. Define in no more than two or three sentences what an "allosteric enzyme" is. (30 points)

5. When a specific threonine near the N-terminus is phosphorylated in yeast glycogen phosphorylase there is a major rearrangement of the 20 N-terminal residues. Twelve residues contract from a beta sheet to a alpha helix. The threonine, formerly bound near the active site, is now bound in a crevice on the other side of the protein. Interestingly, the binding site for the phosphorylated threonine is also the binding site for glucose-6-phosphate, which is a strong allosteric inhibitor of yeast glycogen phosphorylase. Glucose-6-phosphate inhibits the phosphorylated protein cooperatively. When glucose-6-phosphate is bound it displaces the phosphorylated threonine, the N-terminal residues are disordered, and the enzyme is inactivated. Given this information, predict the effect of deleting the N-terminal residues of yeast glycogen phosphorylase. Will the enzyme be active or inactive? Explain your reasoning briefly. (30 points)